

Geometric Feature Extraction of Satellites Based on 3-D Geometry Reconstruction From ISAR Image Sequences

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Extracting the geometric features of satellites from inverse synthetic aperture radar images is a meaningful and challenging task. Previous methods have focused on estimating 3-D geometric features based on their projection mapping relationships with planar features, such as key points, lines, and parallelograms. However, these planar features are unable to fully describe the satellite cabin structure, resulting in limited estimation of only the orientation and length of the cabin's main axis. To extract more diverse features, we propose a framework for geometric feature extraction based on 3-D geometry reconstruction. By reconstructing the complete target structure using target silhouettes extracted from images, our framework can extract a wider range of geometric features of the satellite cabin, including its 3-D shape, 3-D size, and absolute attitude. To ensure precise reconstruction, we design a component completion module to maintain the integrity of the extracted silhouettes and a deviation parameter estimation module to provide the accurate projection parameters. To extract features from the reconstruction results, we propose a 3-D geometric fitting algorithm applicable to different cabin shapes, where the appropriate fitting objective function is selected based on automatic 3-D shape discrimination. The effectiveness of the proposed methods is verified through experiments conducted on both simulated and measured data.

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I. INTRODUCTION

The analysis of the on-orbit state of satellite targets is a promising research area in space situation awareness [1], [2], [3]. Geometric feature extraction [4], [5], [6], [7], [8], [9], [10], 3-D geometry reconstruction [11], [12], [13], and dynamic analysis [14], [15], [16] are core technologies employed in the analysis of on-orbit states. These technologies provide important information for monitoring satellite status, analyzing satellite intentions, and identifying satellite types. Geometric features, such as the absolute attitude and 3-D size, can describe the on-orbit state and geometric characteristics of satellites. These features are typically extracted through the interpretation of high-resolution inverse synthetic aperture radar (ISAR) images obtained via ground-based ISAR [4], [5], [6], [7], [8], [9], [10]. The advancement of high-resolution ISAR imaging techniques has made real-time and accurate feature extraction possible [17], [18], [19], [20], [21], [22], [23].

The cabin is a standard satellite component and possesses geometric features that are crucial for satellite recognition and surveillance. First, the shape and size of the cabin can serve as evidence for satellite target classification [24], [25]. A variety of common shapes, along with their corresponding satellites, are outlined in Table I. Additionally, the orientation of the cabin's main axis can reflect the on-orbit attitude of the satellite, thereby aiding in the analysis of the satellite's status.

In most existing studies, planar features, which are projections of 3-D geometric features, are extracted to describe the target's structural information in ISAR images. The 3-D geometric features are subsequently estimated based on the theory of ISAR imaging projection. Various planar features, such as key points [6], vectors [5], [8], and parallelograms [4], [9], [10], are utilized. For example, in [6], two key points were extracted and linked through the key point extraction network to describe the projection of a cylindrical-like cabin's main axis. Wang et al. [8] employed a vector obtained through 2-D principal component analysis (PCA) to describe the projection of the main axis of a cabin. In [4], the projection of a rectangular component is expressed as a parallelogram, which is extracted using background segmentation and line detection. Subsequently, an optimization function is established based on the mapping relationship between planar features and geometric features [4], [5], [6], [7], [8], [9], [10], [11], [12], enabling the estimation of geometric features through solving the optimization problem. However, in reality, projections of the satellite cabin exhibit complexity and variability, making it difficult for these planar features to fully describe the structural information. For instance, under different observation views, the projection of a cylinder can appear as a circle, a rectangle, or an intricate figure composed of multiple semiellipses and a parallelogram. The loss of structural information in the planar feature description limits the diversity of the extracted geometric features. Consequently, these methods can only estimate the main axis of a cylindrical or cubic cabin [5], [6], [8].

TABLE I
Shapes of Common Satellite Cabins

Shape of satellite cabin	Examples
Sphere	Dongfanghong-1, The Humanity Star, BLITS-M, Transit 2A
Quasi-Cylinder	KH-11, AcrimSAT, ISEE-3, Hubble Space Telescope, Tiangong-2, Tianzhou-1, Transit-3, Solrad-9
Cuboid	OSTM-Jason-2, ICESat, CloudSat, Fox-1, Galaxy-30, SES-20, Beidou-3, Orbicraft-Zorkiy, CubeRRT, GPS-3

As an alternative technique for on-orbit state analysis, 3-D geometry reconstruction can be used to reconstruct the 3-D structure of a target based on the ISAR image sequence. One typical reconstruction method [11] reconstructs the 3-D structure using silhouette information in accordance with imaging projection theory. Silhouettes effectively describe the target's structural information in images, and a direct mapping relationship is established between 3-D structure and 2-D silhouettes. Thus, Zhou et al. [11] can obtain the complete structure of the target with its actual attitude.

In theory, the reconstruction result has the potential to provide more target structural information for geometric feature extraction than the planar features used in the existing methods [27]. As a result, more diverse geometric features can be extracted based on the 3-D geometry reconstruction. However, there are challenges in extracting geometric features from reconstruction results. On the one hand, the accuracy of the reconstruction needs improvement. The radar line of sight (LOS) parameters in the tracking system are not precise enough, leading to silhouettes deviating from the imaging center [11]. A more accurate registration of the target centroid is essential to ensure reconstruction precision. Additionally, occlusion between satellite components in images should be considered because occlusion may result in incomplete reconstruction. On the other hand, the reconstruction results are in the form of point clouds, but the existing point cloud processing technologies are not suitable for extracting cabin features. The 3-D geometric fitting is commonly used to estimate the size and attitude of point clouds. However, the existing fitting algorithms, such as [44] for cuboid fitting and [45] for cylinder fitting, are typically designed for specific geometric shapes. Given that satellite cabins can have diverse and unknown shapes, it is challenging for these fitting methods to automatically differentiate the cabin shape.

In this article, we propose a framework based on 3-D geometry reconstruction to extract more diverse geometric features. First, cabin silhouettes are extracted from images using a specially designed component completion (CC) module to ensure the integrity of the silhouettes. Then, 3-D geometry reconstruction and deviation parameter estimation (DPE) are alternately implemented to enhance the reconstruction precision. Target projections are registered using the estimated deviation parameters during reconstruction, while DPE estimates parameters by

maximizing the number of target points projected into target areas, where the target points are approximated using the current reconstruction result. To extract geometric features, we propose a 3-D geometric fitting algorithm that can be applied to different shapes. Considering that three views of common cabins determined by 3-D PCA are proven to be rectangles or ellipses, we perform 3-D geometric fitting by fitting these three views. In the three-view fitting, we define a rectangle-or-ellipse discrimination operator based on the relationship between the polygon area and vertices, and we design optimization objective functions according to the discriminated shapes. By analyzing the geometric fitting results, we can achieve shape discrimination, 3-D size estimation, and absolute attitude estimation of the satellite cabin. We conducted experiments on three satellites with different shapes to demonstrate the advancements and universality of our framework. The main contributions of our proposed method are as follows.

- 1) We propose a novel framework for geometric feature extraction based on 3-D geometry reconstruction. Our framework can extract a wider range of features than the existing methods. Specifically, for satellite cabins, our framework enables shape discrimination, 3-D size estimation, and absolute attitude estimation, which surpass the limitations of current methods that only estimate the size and direction of the main axis.
- 2) Considering the occlusion between satellite components, we design a CC module to mitigate its impact on 3-D geometry reconstruction. Our module focuses on key geometric properties of the occluded cabins and employs a point-by-point judgment method for CC.
- 3) We improve the 3-D geometry reconstruction accuracy (RA) by registering target centroids among the image sequence. The target centroid deviations are estimated using the alternating iteration strategy and an optimization-based DPE module.
- 4) To the best of our knowledge, this is the first introduction of a 3-D geometric fitting algorithm that is applicable to various shapes, including spheres, cuboids, and cylinders. When dealing with a reconstructed cabin point cloud, our algorithm can discern its 3-D shape and automatically select suitable geometric fitting objective functions.

The rest of this article is organized as follows. In Section II, the radar observation geometry for satellites is introduced. In Section III, a detailed description of our geometric feature extraction framework is provided. In Section IV, several experiments are conducted to verify the validity of our methods. Finally, Section V concludes this article.

II. OBSERVATION GEOMETRY AND IMAGING PROJECTION MODEL

In this section, the observation geometry between the satellite and the ground-based radar is introduced, and the connection between the satellite's 3-D structure and its

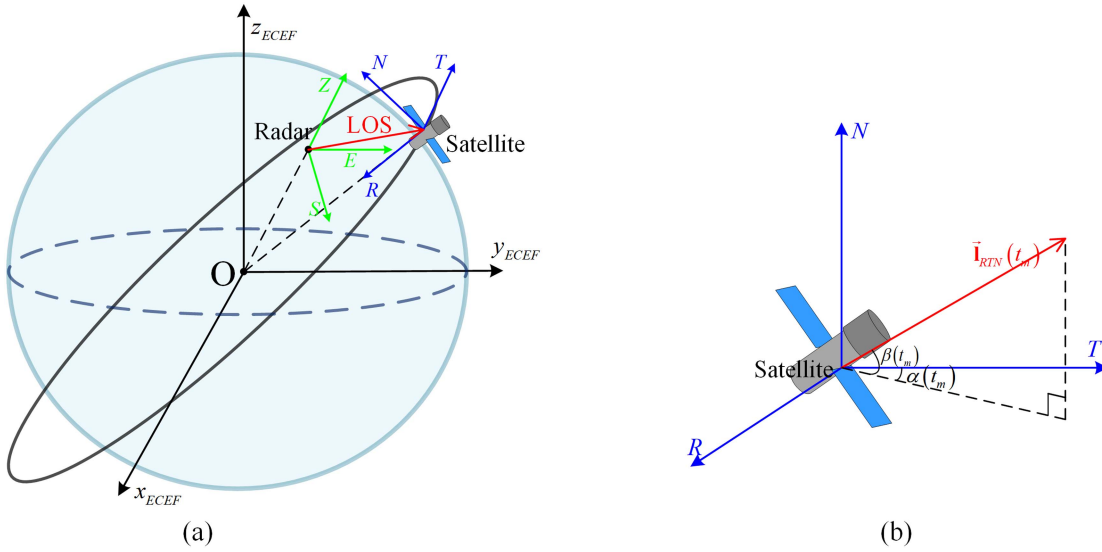


Fig. 1. Diagram of the observation geometry. (a) Radar observation geometry of the satellite in orbit. (b) LOS in the RTN coordinate system.

2-D radar image is established by the imaging projection model [27]. As shown in Fig. 1, the satellite is assumed to be observed in an orbital coordinate system named RTN system, whose axes are defined by the radial (R), along-track (T), and cross-track (N) directions. Many satellites can be deemed three-axis stable during the observation, and their attitudes remain unchanged in the RTN system under the attitude controller [28]. Therefore, we can describe the relative rotational motion between the satellite and radar using the variations in the LOS in the RTN system [4], [5], [6], [7], [8], [9], [10].

In practice, the instantaneous LOS vectors in the RTN system are obtained from the LOS in the site-centric [South-East-Z (SEZ)] coordinate system by coordinate transformation

$$\vec{I}_{RTN}(t_m) = \mathbf{R}_{SEZ}^{RTN} \vec{I}_{SEZ}(t_m) \quad (1)$$

where $\vec{I}_{RTN}(t_m), \vec{I}_{SEZ}(t_m) \in \mathbb{R}_{3 \times 1}$ are, respectively, the LOS vectors in the RTN and SEZ system, $\vec{I}_{SEZ}(t_m)$ are provided by the radar tracking system; t_m is the slow time in an imaging coherent processing interval (CPI), $m = 1, 2, \dots, M$, and M is the pulse number. The coordinate transformation matrix $\mathbf{R}_{SEZ}^{RTN}$ is determined based on the Kepler two-body motion property [28].

As shown in Fig. 2, the ISAR imaging process can be understood as projecting a stable attitude target onto an imaging plane, which is determined by the range direction and the azimuth direction [27]. Based on this theory, for a scattering point $\vec{p} = [x, y, z]^T$ in the RTN system, its projection position $[r, d]^T$ in the range-doppler (RD) image can be calculated with the imaging projection matrix $\mathbf{H}$ as follows:

$$\begin{bmatrix} r \\ d \end{bmatrix} = \mathbf{H} \vec{p} + \begin{bmatrix} d_r \\ d_a \end{bmatrix} \quad (2)$$

$$\mathbf{H} = \begin{bmatrix} \vec{r}^T / \Delta r \\ \vec{f}_d^T / \Delta f_d \end{bmatrix}_{2 \times 3}$$

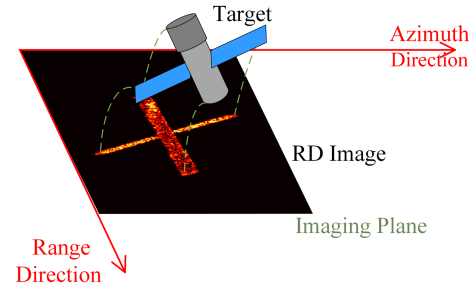


Fig. 2. ISAR imaging projection model.

where d_r and d_a are, respectively, the deviation parameters in the range and azimuth directions, which are defined as the distances between the target centroid and imaging center. When the imaging center is registered with the coordinate origin of the RTN system, the deviation parameters equal zero. $\vec{r}$ and $\vec{f}_d$ are, respectively, the range and azimuth directions

$$\begin{cases} \vec{r} = \vec{I}_{RTN}(t_c) \\ \vec{f}_d = \frac{\partial \vec{I}_{RTN}(t_c)}{\partial t_m} \end{cases} \quad (3)$$

where t_c is the central moment of the CPI, and $\partial \vec{I}_{RTN}(t_c) / \partial t_m$ represents the partial derivative of $\vec{I}_{RTN}(t_m)$ at t_c . Δr and Δf_d are, respectively, the range and cross-range resolution

$$\begin{cases} \Delta r = \frac{c}{2B} \\ \Delta f_d = \frac{\lambda}{2\Omega} \end{cases} \quad (4)$$

where c is the light speed, B is the signal bandwidth, λ is the signal wavelength, and Ω is the relative rotation angle during the CPI. For the three-axis stabilized satellites, Ω is the angle between $\vec{I}_{RTN}(t_1)$ and $\vec{I}_{RTN}(t_M)$ [30].

The imaging projection model described by (2) makes it possible to rebuild the complete target structure from images [4], [5], [6], [7], [8], [9], [10].

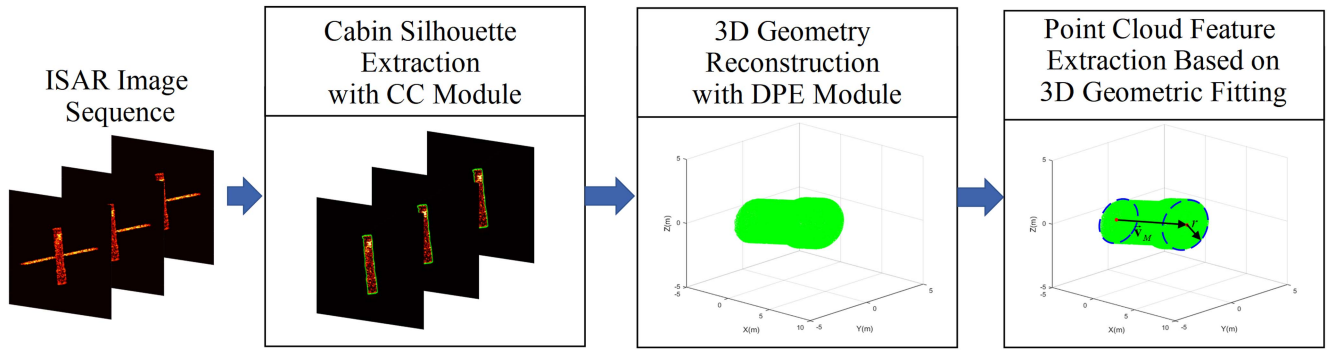


Fig. 3. Flowchart of the proposed geometric feature extraction framework.

III. GEOMETRIC FEATURE EXTRACTION BASED ON 3-D GEOMETRY RECONSTRUCTION

A. Flowchart of the Proposed Method

As shown in Fig. 3, our geometric feature extraction, which consists of three essential steps, is introduced in this section. First, cabin silhouettes are extracted from the ISAR image sequence, with a CC module designed to handle the occluded parts. Then, the cabin's 3-D structure is rebuilt from the extracted silhouettes together with the DPE. Finally, a 3-D geometric fitting algorithm applicable to different shapes is proposed, and geometric features of the satellite cabin, including its 3-D shape, 3-D size, and absolute attitude, are obtained from the fitting result. The details of the steps are shown in the subsequent sections.

B. Cabin Silhouette Extraction

To extract cabin silhouettes from ISAR images, it is necessary to segment the cabin pixels. The existing works on satellite geometric feature extraction [5], [6], [7], [8] have combined component segmentation and background segmentation for cabin pixel extraction. Semantic segmentation methods, such as the FCN-based method [31], the U-Net-based method [32], and the Pix2pixGAN-based method [8], are commonly used for component segmentation, while the constant false alarm rate algorithm [34], which can remove pixels of noise and sidelobes, is employed for background segmentation.

However, under certain observation views, cabins in images may be partially covered by other satellite components. The occluded parts vary with observation angles, making it challenging for these data-driven segmentation methods to accurately complete the occluded parts.

To ensure the integrity of the extracted cabin silhouettes, we design a CC module according to the following two characteristics of the occluded parts.

- 1) The covered part of the cabin is composed of the target pixels.
- 2) Without occlusion, the projection of a typical cabin on the imaging plane resembles a convex polygon.

As shown in Fig. 4, satellite pixels extracted by background segmentation and the marked cabin regions labeled

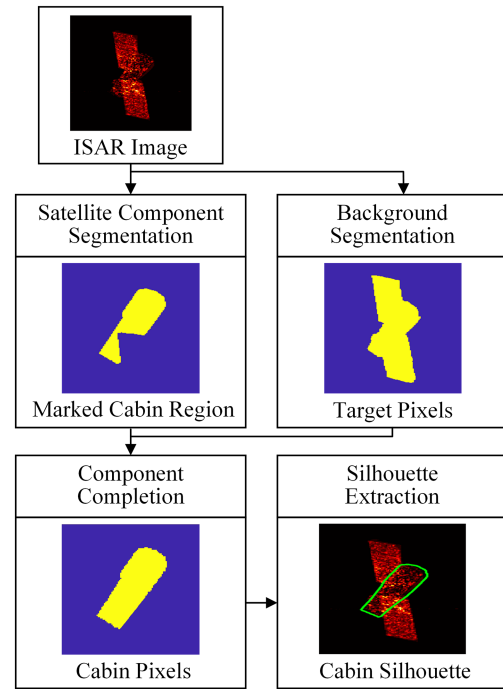


Fig. 4. Flowchart of the cabin silhouette extraction.

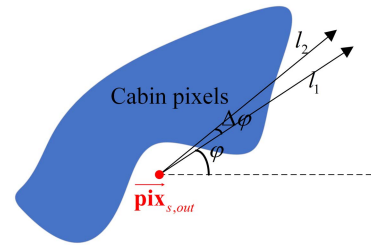


Fig. 5. Diagram of the two rays.

by component segmentation are utilized in our CC. Our CC is achieved by pointwise judgment: for each satellite pixel $\vec{\text{pix}}_{s,\text{out}} \in \mathbb{R}_{2 \times 1}$, which is located outside of the marked cabin region, its judgment procedure is given in Algorithm 1. In judgment, the relationship between the pixel and two rays is shown in Fig. 5

After the judgment, the binary image of the complete cabin can be generated. The cabin silhouette Sil (labeled

Algorithm 1: Pointwise Judgment for CC.

- 1: Input:** The pixel $\vec{\text{pix}}_{s,\text{out}}$. The cabin pixels $\{\vec{\text{pix}}_{c,i}\}$, $i = 1, 2, \dots, N_{\text{cabin}}$, N_{cabin} is the number of pixels in the marked cabin regions. The search interval $\Delta\varphi$ ($\Delta\varphi > 0$).
 - 2: Output:** The judgment result: *Cabin* or *other components*.
-
- 3: Initialize:** $\varphi = 0$, $Ps = 0$
where φ is the bevel angle of a ray, and Ps is the probability that $\vec{\text{pix}}_{s,\text{out}}$ belongs to the cabin.
 - 4: While** $\varphi \leq 360^\circ$ **do:**
 - 4.1** Calculate the number N_{cp} of cabin pixels located between two rays. As shown in Fig. 5, the vertex of rays is $\vec{\text{pix}}_{s,\text{out}}$, and their bevel angles are φ and $\varphi + \Delta\varphi$.
 - 4.2** Update the probability

$$Ps = Ps + \frac{\text{sgn}(N_{cp})}{\lfloor 360/\Delta\varphi \rfloor}$$
where $\text{sgn}(\cdot)$ is the sign function, and $\lfloor \cdot \rfloor$ is the floor function.
 - 4.3** $\varphi = \varphi + \Delta\varphi$.
 - 5: Judge:** If $Ps > 0.5$, $\vec{\text{pix}}_{s,\text{out}}$ belongs to the cabin.
Otherwise, $\vec{\text{pix}}_{s,\text{out}}$ belongs to other components.

green in Fig. 4), which is composed of several side points arranged clockwise, is further extracted from the cabin binary image by boundary extraction methods, such as morphological operations [35].

C. Three-Dimensional Geometry Reconstruction for the Cabin

Based on ISAR imaging projection theory, we can rebuild the target 3-D structure from the extracted silhouettes. In reconstruction, target points are selected from a predefined 3-D candidate point set by determining whether their projections fall within the target silhouettes [11]. Typically, the candidate point set Cset_0 is set as a cube whose side length equals the farthest distance between target pixels in an image. Although this method can rebuild the complete target structure with the actual attitude, its accuracy should be further improved [11].

In practice, the position of the target centroid, which is estimated by LOS parameters from the tracking system [36], is taken as the RD imaging center. However, the LOS parameters are not precise enough to avoid the actual target centroid deviating from the imaging center, resulting in some target points being projected out of the target area in the ISAR image.

To estimate the centroid deviations and improve the RA, we introduce the alternating iteration strategy into the reconstruction, and its flowchart is given in Fig. 6.

In Fig. 6, N is the total number of images. For the n th image ($n = 1, 2, \dots, N$), the silhouette Sil_n is extracted in Section III-B, $\mathbf{H}_n$ is the imaging projection matrix, $d_{r,n}^{(T)}$

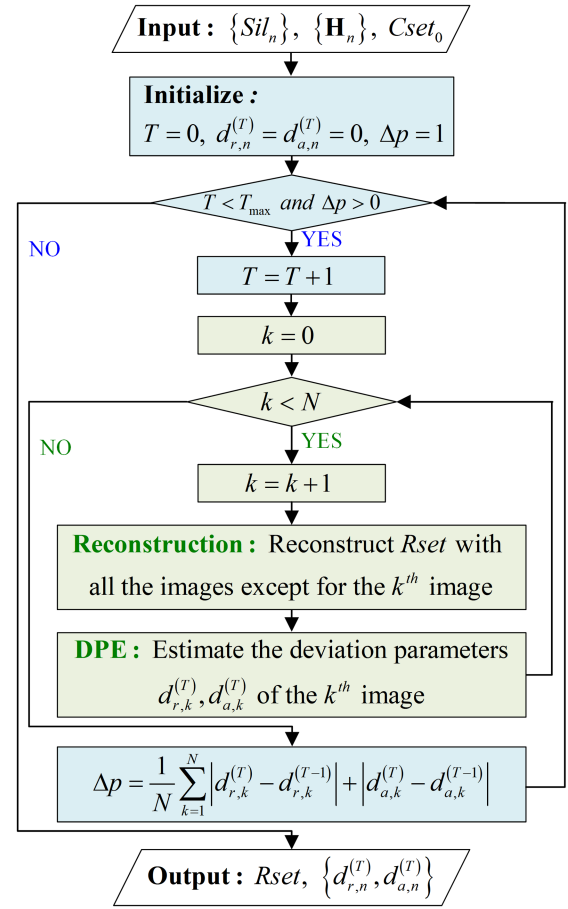


Fig. 6. Flowchart of 3-D geometry reconstruction with DPE.

and $d_{a,n}^{(T)}$ are the deviation parameters estimated in the T th iteration, Δp is the average change of deviation parameters in an iteration, and $T_{\text{max}} = 10$ is the maximum number of iterations. The parts marked green illustrate that the reconstruction and DPE are alternatively implemented.

- 1) The target point cloud Rset is reconstructed by the method in [11], whose details are provided in Algorithm II. In reconstruction, we correct projections of candidate points $\vec{\text{p}}_{c,i}$ with the deviation parameters $\{d_{r,n}^{(T-1)}, d_{a,n}^{(T-1)}\}$ estimated in the last iteration.
- 2) The deviation parameters $\{d_{r,k}^{(T)}, d_{a,k}^{(T)}\}$ are estimated by maximizing the number of target points projected into the target areas, where target points are approximated with the current reconstruction result Rset . The estimation process is formulated as follows:

$$d_{r,k}^{(T)}, d_{a,k}^{(T)} = \arg \max_{d_r, d_a} \sum_{i=1}^{N_R} \text{Jud}(\vec{\text{pro}}_{i,k}(d_r, d_a), \text{Sil}_k)$$

$$\vec{\text{pro}}_{i,k}(d_r, d_a) = \mathbf{H}_k \vec{\text{p}}_{r,i} + \begin{bmatrix} d_r \\ d_a \end{bmatrix} \quad (5)$$

where $\vec{\text{p}}_{r,i} \in \mathbb{R}_{3 \times 1}$ is the i th point in Rset , $i = 1, 2, \dots, N_R$, where N_R is the number of points in Rset . $\vec{\text{pro}}_{i,k}(d_r, d_a)$ is the projection of $\vec{\text{p}}_{r,i}$ in the k th image, and $\text{Jud}(\cdot)$ is the judgment function defined in Algorithm 2.

Algorithm 2: 3-D Geometry Reconstruction.

1: Input: The silhouettes $\{\text{Sil}_n\}$, the imaging projection matrices $\{\mathbf{H}_n\}$, the deviation parameters $\{d_{r,n}^{(T-1)}, d_{a,n}^{(T-1)}\}$, and the 3-D candidate point set Cset_0 . $n = 1, 2, \dots, N$, and $n \neq k$.

2: Output: The reconstructed target point cloud Rset .

3: Calculate projections of the candidate points

$$\vec{\text{pro}}_{i,n} = \mathbf{H}_n \vec{\text{p}}_{c,i} + \begin{bmatrix} d_{r,n}^{(T-1)} \\ d_{a,n}^{(T-1)} \end{bmatrix}$$

where $\vec{\text{p}}_{c,i} \in \mathbb{R}_{3 \times 1}$ is the i th point in Cset_0 , $i = 1, 2, \dots, N_C$, N_C is the number of points in Cset_0 ; $\vec{\text{pro}}_{i,n}$ is the projection of $\vec{\text{p}}_{c,i}$ in the n th image.

4: Judge whether the projections locate in target areas enclosed by the silhouettes, and calculate the voting scores for the candidate points

$$S(\vec{\text{p}}_{c,i}) = \frac{1}{N-1} \sum_{n \neq k} \text{Jud}(\vec{\text{pro}}_{i,n}, \text{Sil}_n)$$

where $S(\vec{\text{p}}_{c,i})$ is the voting score for $\vec{\text{p}}_{c,i}$. The judgment function $\text{Jud}(\cdot)$ equals 1 when $\vec{\text{pro}}_{i,n}$ locates in the target area enclosed by Sil_n ; otherwise, $\text{Jud}(\cdot)$ equals 0.

5: Construct Rset with the points that satisfy $S(\vec{\text{p}}_{c,i}) \geq \xi$, $\xi = 0.9$ is a preset pass rate.

Considering that the target is located near the imaging center after motion compensation [36], the deviation parameters d_r and d_a can be regarded as integers close to zero. Then, the optimization (5) can be solved by traversing methods rapidly.

D. Geometric Feature Extraction Based on 3-D Geometric Fitting

In this section, our 3-D geometric fitting algorithm applicable to different cabin shapes is introduced. Considering the diversity of cabin shapes, we cannot directly model the fitting process as an integrated optimization problem, as in the existing fitting methods [44], [45], [46]. To address this challenge, we reveal the properties of 3-D PCA on a reflective symmetric point cloud. By 3-D PCA, we can decompose 3-D geometric fitting into three-view fitting. In three-view fitting, a rectangle-or-ellipse discrimination operator is defined to guide the design of the geometric fitting objective functions. Finally, by comprehensively utilizing the PCA vectors and three-view fitting results, the 3-D shape, size, and attitude of the cabin can be obtained. The procedure of geometric feature extraction based on 3-D geometric fitting is shown in Fig. 7.

1) *Three Views Extraction:* The shapes of common satellite cabins, including cylinders, cuboids, and spheres, are reflective symmetric. Fig. 8 illustrates the reflective symmetry planes of these cabins, denoted by red, green, and black dotted lines. Three views of these cabins are rectangles or ellipses when normal vectors of their reflective symmetry planes are selected as the projection directions.

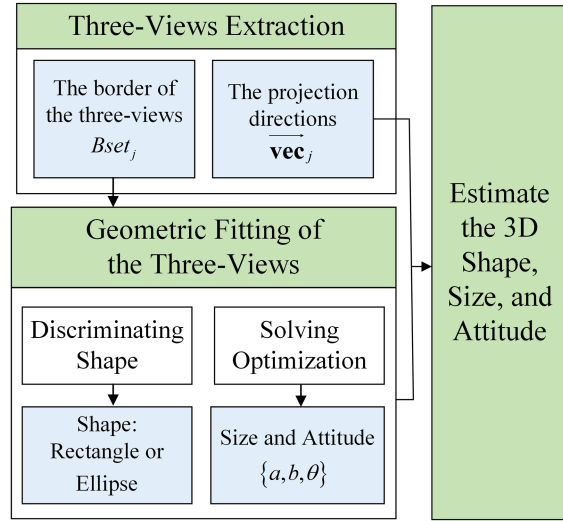


Fig. 7. Geometric feature extraction based on 3-D geometric fitting.

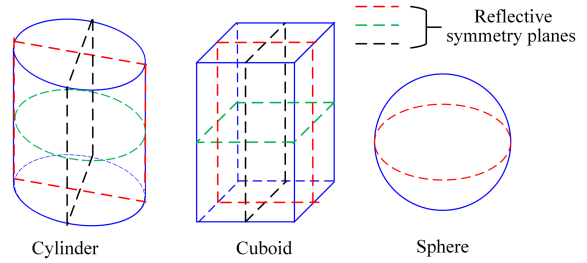


Fig. 8. Reflective symmetry planes of cylinders, cuboids, and spheres.

Therefore, 3-D geometric fitting can be realized by fitting the specific three views if the normal vectors of the reflective symmetry planes are estimated.

The normal vector of the reflective symmetry plane is proven to be available by 3-D PCA. Details of the proof are given in Appendix I. Based on this property, we extract three views of the cabin point cloud via 3-D PCA. The details are as follows.

Step 1: Calculate the covariance matrix of the reconstructed cabin point cloud Rset

$$\mathbf{C} = \frac{1}{N_R} \bar{\mathbf{W}} \bar{\mathbf{W}}^T$$

$$\bar{\mathbf{W}} = [\vec{\text{p}}_{cr,1}, \dots, \vec{\text{p}}_{cr,i}, \dots, \vec{\text{p}}_{cr,N_R}]_{3 \times N_R} \quad (6)$$

where $\{\vec{\text{p}}_{cr,i}\}$ is obtained by centralizing Rset : $\vec{\text{p}}_{cr,i} = \vec{\text{p}}_{r,i} - \frac{1}{N_R} \sum_{i=1}^{N_R} \vec{\text{p}}_{r,i}$, $\vec{\text{p}}_{r,i} \in \text{Rset}$.

Step 2: Calculate three eigenvectors $\vec{\text{vec}}_j \in \mathbb{R}_{3 \times 1}$ ($j = 1, 2, 3$) of the covariance matrix $\mathbf{C}$ via eigenvalue decomposition.

Step 3: Calculate projection matrices $\mathbf{H}_{P,j}$ ($j = 1, 2, 3$) of three views with the eigenvectors $\vec{\text{vec}}_j$

$$\mathbf{H}_{P,1} = [\vec{\text{vec}}_1^T; \vec{\text{vec}}_2^T]_{2 \times 3}$$

$$\mathbf{H}_{P,2} = [\vec{\text{vec}}_2^T; \vec{\text{vec}}_3^T]_{2 \times 3}$$

$$\mathbf{H}_{P,3} = [\vec{\text{vec}}_3^T; \vec{\text{vec}}_1^T]_{2 \times 3}. \quad (7)$$

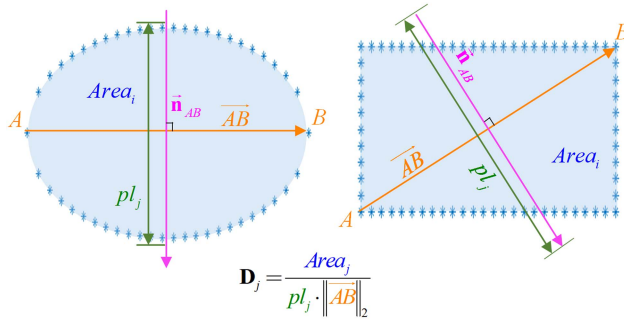


Fig. 9. Diagram of the shape discrimination operator.

Step 4: Project $\{\vec{p}_{cr,i}\}$ onto the planes constructed by eigenvectors

$$\text{Tset}_j = \left\{ \vec{tp}_{j,i} \mid \vec{tp}_{j,i} = \mathbf{H}_{P,j} \vec{p}_{cr,i} \right\} \quad (8)$$

where Tset_j are the three-view point sets, and $\vec{tp}_{j,i} \in \mathbb{R}_{2 \times 1}$ is the i th point in Tset_j , $i = 1, 2, \dots, N_R$.

Step 5: The shape of the three views can be described by Tset_j 's border. Since our CC module utilizes the convexity of cabin projections, the shapes of the three views are convex. Then, its border point set Bset_j can be solved by convex hull algorithms [37], [38], [39], [40].

2) *Geometric Fitting of the Three Views:* As shown in Fig. 7, the three-view fitting is divided into two steps: discriminating shape and solving optimization. Since the cabin's three views can be a rectangle or an ellipse, we need to discriminate the three-view shapes to guide the design of the optimization objective functions.

a) *Discriminating shape* To discriminate shapes accurately and quickly, we design a novel shape discrimination operator based on the relationship between vertex positions and geometric area. The shape discrimination operator can be calculated as follows.

Step 1: Find the two vertices A and B farthest apart in Bset_j .

Step 2: Calculate the normal vector $\vec{n}_{AB}$ of $\vec{AB}$. $\vec{n}_{AB}, \vec{AB} \in \mathbb{R}_{2 \times 1}$.

Step 3: Project the points $\vec{bp}_{j,i} \in \text{Bset}_j$ onto $\vec{n}_{AB}$ and calculate the projection length pl_j

$$pl_j = \max_i \left(\vec{n}_{AB}^T \cdot \vec{bp}_{j,i} \right) - \min_i \left(\vec{n}_{AB}^T \cdot \vec{bp}_{j,i} \right). \quad (9)$$

Step 4: Calculate the area Area_j of the region enclosed by the points in Bset_j .

Step 5: Calculate the shape discrimination operator $\mathbf{D}_j$ as follows:

$$\mathbf{D}_j = \frac{\text{Area}_j}{pl_j \cdot \|\vec{AB}\|_2} \quad (10)$$

where $\|\cdot\|_2$ represents the norm of a vector.

Step 6: Determine the shape: if $\mathbf{D}_j < \text{thr}_D$, the three-view image is regarded as a rectangle; otherwise, it is regarded as an ellipse. thr_D is a manual threshold.

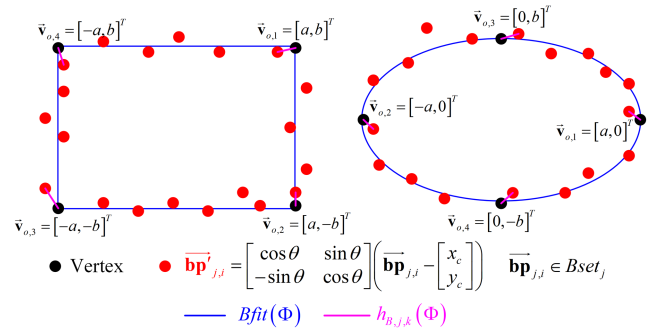


Fig. 10. Diagram of the rectangular fitting and the elliptical fitting methods.

As shown in Fig. 9, our operator is independent of the target size, which illustrates that our operator is effective.

- 1) For a rectangle, $\vec{AB}$ is its diagonal, and the area is $pl_j \cdot \|\vec{AB}\|_2 / 2$. Our operator equals 0.5.
- 2) For an ellipse, $\vec{AB}$ is its major axis, and the area is $\pi \cdot pl_j \cdot \|\vec{AB}\|_2 / 4$. Our operator equals $\pi / 4$.

Therefore, the threshold thr_D can be set as $\frac{\pi/4+0.5}{2}$.

b) *Solving optimization:* The optimization objective function for rectangular or elliptical fitting is designed based on the Euclidean distance. The objective function for the three-view border Bset_j is designed as follows:

$$J_j(\Phi) = \frac{1}{N_{B,j}} \sum_{i=1}^{N_{B,j}} h_{A,j,i}(\Phi) + \frac{1}{4} \sum_{i=1}^4 h_{B,j,k}(\Phi) \quad (11)$$

where Φ is the parameter set of the fitted geometric figure $\text{Bfit}(\Phi)$, and $N_{B,j}$ is the number of points in Bset_j . For the i th point $\vec{bp}_{j,i}$ in Bset_j , $h_{A,j,i}(\Phi)$ is its minimum distance from $\text{Bfit}(\Phi)$. For the k th vertex of $\text{Bfit}(\Phi)$, $h_{B,j,k}(\Phi)$ is its minimum distance from Bset_j .

The calculations of the objective function for rectangle fitting and ellipse fitting are different.

For rectangles, $\Phi = \{a, b, \theta, x_c, y_c\}$. $[x_c, y_c]^T$ is the location of the rectangle centroid, $2a$ and $2b$ are the lengths of the rectangle sides, and θ is the inclination angle between the horizontal axis and the longer side. For the three-view Bset_j , its horizontal axis is $\vec{vec}_j$. We define the centralization of $\vec{bp}_{j,i}$ as follows:

$$\vec{bp}'_{j,i} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \left(\vec{bp}_{j,i} - \begin{bmatrix} x_c \\ y_c \end{bmatrix} \right). \quad (12)$$

As shown in Fig. 10, the $h_{A,j,i}(\Phi)$ is calculated by (13) when $|\vec{bp}'_{j,i}(1)| < a$ and $|\vec{bp}'_{j,i}(2)| < b$; otherwise, it is calculated by (14)

$$h_{A,j,i}(\Phi) = \min(|\text{dbn}_{j,i}(1)|, |\text{dbn}_{j,i}(2)|) \quad (13)$$

$$h_{A,j,i}(\Phi) = \left\| \begin{bmatrix} \max(\text{dbn}_{j,i}(1), 0) \\ \max(\text{dbn}_{j,i}(2), 0) \end{bmatrix} \right\|_2 \quad (14)$$

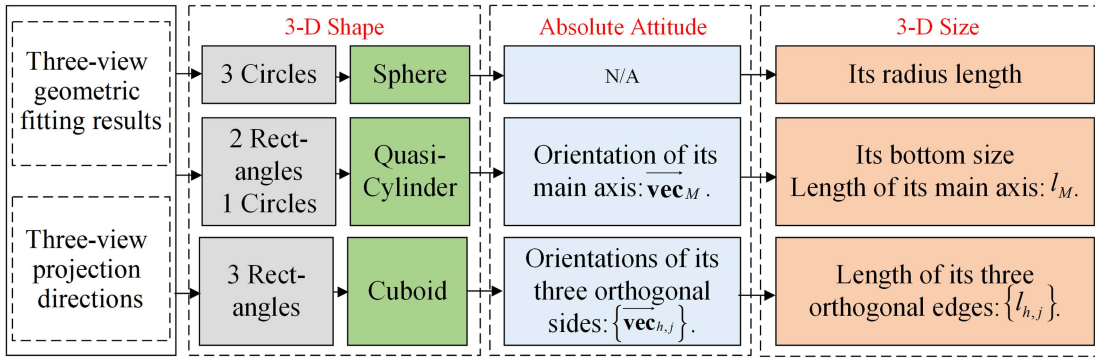


Fig. 11. Estimation of the 3-D shape, size, and attitude.

where $\mathbf{dbn}_{j,i} = |\vec{\mathbf{bp}}'_{j,i}| - [a, b]^T$, and $|\cdot|$ is the absolute value function. The $h_{B,j,k}(\Phi)$ is calculated as follows:

$$h_{B,j,k}(\Phi) = \min_i \left\| \vec{\mathbf{v}}_{o,k} - \vec{\mathbf{bp}}'_{j,i} \right\|_2^2. \quad (15)$$

For rectangles, $\vec{\mathbf{v}}_{o,1} = [a, b]^T$, $\vec{\mathbf{v}}_{o,2} = [a, -b]^T$, $\vec{\mathbf{v}}_{o,3} = [-a, -b]^T$, and $\vec{\mathbf{v}}_{o,4} = [-a, b]^T$.

For ellipses, $\Phi = \{a, b, \theta, x_c, y_c\}$. $[x_c, y_c]^T$ is the centroid location, $2a$ and $2b$ are the lengths of the long axis and short axis, respectively, and θ is the inclination angle between the horizontal axis and the long axis. As shown in Fig. 10, the $h_{A,j,i}(\Phi)$ is calculated by solving the optimization as follows:

$$\begin{aligned} \min_{x,y} & \sqrt{(x - \vec{\mathbf{bp}}'_{j,i}(1))^2 + (y - \vec{\mathbf{bp}}'_{j,i}(2))^2} \\ \text{s.t. } & \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1. \end{aligned} \quad (16)$$

Its optimal solution can be obtained by the Lagrange multiplier. The details are given in Appendix II. The $h_{B,j,k}(\Phi)$ is also calculated by (15). For ellipses, $\vec{\mathbf{v}}_{o,1} = [a, 0]^T$, $\vec{\mathbf{v}}_{o,2} = [-a, 0]^T$, $\vec{\mathbf{v}}_{o,3} = [0, b]^T$, and $\vec{\mathbf{v}}_{o,4} = [0, -b]^T$.

To estimate the location, size, and attitude parameters of the best fit rectangle or ellipse, we minimize the objective function (11) by the particle swarm optimization algorithm [41].

3) *Estimate the 3-D Shape, Size, and Attitude:* As shown in Fig. 11, the 3-D shape, size, and attitude of the cabin can be estimated with the best fit geometric figures and projection directions of the three views. First, we determine the 3-D shape by shapes of the three views. Then, the size and attitude can be estimated as follows.

For a sphere, its size is determined by its radius, which is the mean value of the three-view radii. However, it is impossible for methods based on image interpretation to estimate the attitude of a spherical cabin.

For a quasi-cylinder, its attitude is determined by its main axis orientation $\vec{\mathbf{vec}}_M$, and its size is determined by its main axis length l_M and its bottom size. The bottom size is determined by the long axis and short axis of the

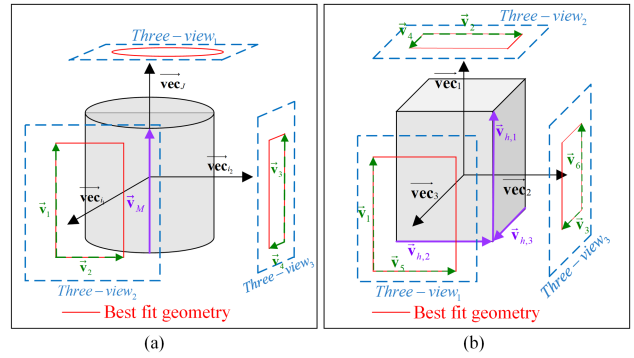


Fig. 12. Diagram of the 3-D attitude and size estimation process. (a) Estimation for a quasi-cylinder. (b) Estimation for a cuboid.

best fit ellipse. As shown in Fig. 12(a), l_M and $\vec{\mathbf{vec}}_M$ can be calculated as follows:

$$\begin{aligned} \vec{\mathbf{v}}_M &= \frac{1}{2} \left([\vec{\mathbf{vec}}_{i_1}, \vec{\mathbf{vec}}_J]_{3 \times 2} \vec{\mathbf{v}}_1 + [\vec{\mathbf{vec}}_{i_2}, \vec{\mathbf{vec}}_J]_{3 \times 2} \vec{\mathbf{v}}_3 \right) \\ l_M &= \|\vec{\mathbf{v}}_M\|_2, \quad \vec{\mathbf{vec}}_M = \vec{\mathbf{v}}_M / l_M \end{aligned} \quad (17)$$

where $\vec{\mathbf{vec}}_J \in \mathbb{R}_{3 \times 1}$ is the normal vector of the ellipse obtained by PCA, and $\vec{\mathbf{vec}}_{i_1}, \vec{\mathbf{vec}}_{i_2} \in \mathbb{R}_{3 \times 1}$ are the other two eigenvectors. $\vec{\mathbf{v}}_1, \vec{\mathbf{v}}_2 \in \mathbb{R}_{2 \times 1}$ are the rectangular sides on $P(\vec{\mathbf{vec}}_{i_1}, \vec{\mathbf{vec}}_J)$, and $\theta(\vec{\mathbf{v}}_1, \vec{\mathbf{vec}}_J) < 45^\circ$, where $P(\cdot)$ represents the plane determined by two vectors and $\theta(\cdot)$ represents the included angle between two vectors. $\vec{\mathbf{v}}_3, \vec{\mathbf{v}}_4 \in \mathbb{R}_{2 \times 1}$ are the rectangular sides on $P(\vec{\mathbf{vec}}_{i_2}, \vec{\mathbf{vec}}_J)$, and $\theta(\vec{\mathbf{v}}_3, \vec{\mathbf{vec}}_J) < 45^\circ$. Ideally, the included angles among $\vec{\mathbf{v}}_1, \vec{\mathbf{v}}_3$, and $\vec{\mathbf{vec}}_J$ are zero; then, $\vec{\mathbf{vec}}_M = \vec{\mathbf{vec}}_J$.

For a cuboid, its attitude is determined by the orientations of its three orthogonal sides $\vec{\mathbf{vec}}_{h,j}$ ($j = 1, 2, 3$), and its size is determined by the side lengths $l_{h,j}$. As shown in Fig. 12(b), $\vec{\mathbf{v}}_{h,j} = l_{h,j} \cdot \vec{\mathbf{vec}}_{h,j}$ can be calculated as follows:

$$\begin{aligned} \vec{\mathbf{v}}_{h,1} &= \frac{1}{2} \left([\vec{\mathbf{vec}}_1, \vec{\mathbf{vec}}_2]_{3 \times 2} \vec{\mathbf{v}}_1 + [\vec{\mathbf{vec}}_3, \vec{\mathbf{vec}}_1]_{3 \times 2} \vec{\mathbf{v}}_6 \right) \\ \vec{\mathbf{v}}_{h,2} &= \frac{1}{2} \left([\vec{\mathbf{vec}}_1, \vec{\mathbf{vec}}_2]_{3 \times 2} \vec{\mathbf{v}}_5 + [\vec{\mathbf{vec}}_2, \vec{\mathbf{vec}}_3]_{3 \times 2} \vec{\mathbf{v}}_2 \right) \\ \vec{\mathbf{v}}_{h,3} &= \frac{1}{2} \left([\vec{\mathbf{vec}}_2, \vec{\mathbf{vec}}_3]_{3 \times 2} \vec{\mathbf{v}}_4 + [\vec{\mathbf{vec}}_3, \vec{\mathbf{vec}}_1]_{3 \times 2} \vec{\mathbf{v}}_3 \right) \end{aligned} \quad (18)$$

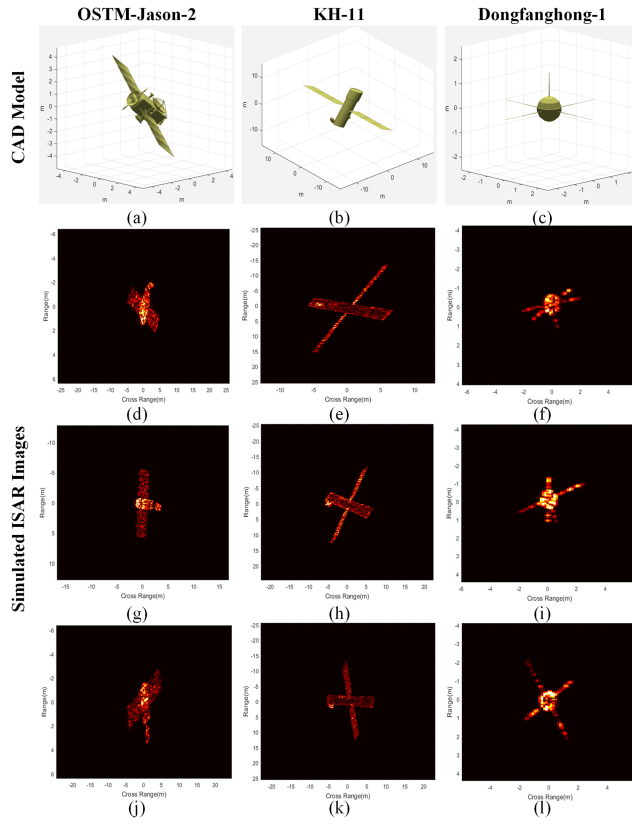


Fig. 13. CAD models and simulated ISAR images of the satellites. (a)–(c) Target CAD models. (d)–(i) Simulated ISAR images.

TABLE II
Radar Location and Parameters

Radar Location	Hangzhou(120.2E,30.3N,0m)	
Radar Parameters	Center Frequency	16.67GHz
	PRF	25Hz
	Bandwidth	4GHz

where $\vec{v}_1, \vec{v}_5 \in \mathbb{R}_{2 \times 1}$ are the rectangular sides on $P(\vec{vec}_1, \vec{vec}_2)$, $\vec{v}_2, \vec{v}_4 \in \mathbb{R}_{2 \times 1}$ are the rectangular sides on $P(\vec{vec}_2, \vec{vec}_3)$, $\vec{v}_3, \vec{v}_6 \in \mathbb{R}_{2 \times 1}$ are the rectangular sides on $P(\vec{vec}_3, \vec{vec}_1)$, and $\theta(\vec{v}_{i_n}, \vec{vec}_j) < 45^\circ$, where the subscript i_n equals j . In ideal status, $\vec{vec}_{h,j} = \vec{vec}_j$.

IV. EXPERIMENTAL ANALYSIS

A. Image Sequences Acquisition

Considering the lack of measured data for satellites with different cabin shapes, we generate simulation data for three satellites via the physical optics (PO) algorithm [42]. As shown in the first row of Fig. 13, computer-aided design (CAD) models of the satellites, including OSTM-Jason-2 (cubic cabin), KH-11 (cylindrical cabin), and Dongfanghong-1 (spherical cabin), are available.¹ In our experiments, a Ku-band radar, whose parameters are listed in Table II, is adopted to generate radar echoes, and the satellites are considered to be three-axis stabilized targets.

¹[Online]. Available: nasa3d.arc.nasa.gov/models

TABLE III
Number of the Used ISAR Image

Satellite	OSTM-Jason-2	KH-11	Dongfanghong-1
Image Number	60	60	45

For each target, three-lap observation angles in the SEZ system are generated with the radar location and satellite orbital elements [4]. The corresponding observation angles in the RTN system, whose variation curves are shown in Fig. 14, are further calculated via coordinate transformation [4]. Subsequently, radar echoes for each target are generated by the PO algorithm based on the observation angles and the target CAD models [42]. Well-focused and scaled ISAR images are generated from the echoes by RD imaging algorithms [17], [18], [19], [20], [21], with a CPI of 10.24 s and an imaging interval of 25 s. The numbers of generated images are shown in Table III, and some simulated ISAR images are shown in the second to fourth rows of Fig. 13.

To highlight our contributions, several experiments have been implemented. In Section Section IV-B, the performance of the improved 3-D geometry reconstruction algorithm is analyzed. In Section IV-C, our 3-D geometry fitting accuracy is proven via Monte Carlo experiments. In Section IV-D, the 2-D PCA-based method [8] is taken as a comparison method to show the accuracy and wide applicability of our geometric feature extraction framework.

B. Performance of the Improved 3-D Geometry Reconstruction Algorithm

In this part, we evaluate the performance of our 3-D geometry reconstruction algorithm. The satellite and cabin reconstruction results are given in Fig. 15, where the first and third rows show the real point clouds, and the second and fourth rows show the reconstructed point clouds. As shown in Fig. 14, the target 3-D structure and absolute attitude can be accurately reflected by our reconstructed point clouds.

To further illustrate the feasibility and effectiveness of our reconstruction algorithm, we conduct the following experimental analysis. First, we examine the convergence of our algorithm. Second, we evaluate the accuracy of DPE. Third, we assess the reliability of CC. Additionally, improvements brought by DPE and CC are verified through an ablation study. Finally, we analyze the computational complexity of our algorithm.

1) *Convergence of the Reconstruction With DPE:* We realize the reconstruction with all the generated images, and the convergence curves of the algorithm on three satellites are shown in Fig. 16. The x -axis is the number of iterations, and the y -axis is the average change Δp of the estimated deviation parameters, which can be calculated by the corresponding equation in Fig. 6. For all three satellites, the average change Δp becomes zero after five iterations, indicating convergence.

2) *Accuracy of the DPE:* Some target registration results are shown in Fig. 17. The images in the first column are the ISAR images and target projection results without

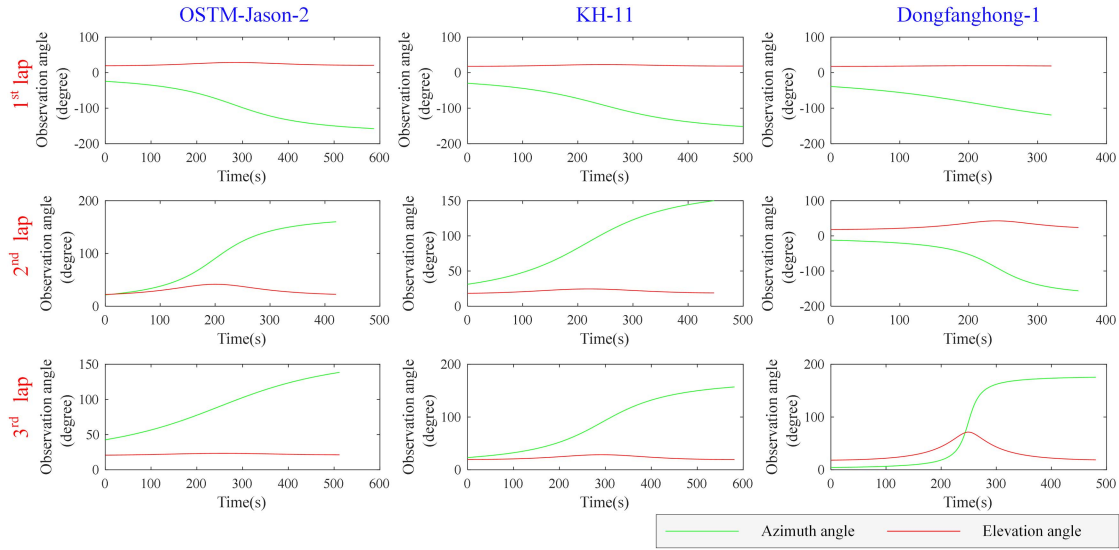


Fig. 14. Variation curves of the generated observation angles. Observation angles of the same target are in the same column; observation angles in the same lap are in the same row; green curves are the azimuth angles; and red curves are the elevation angles.

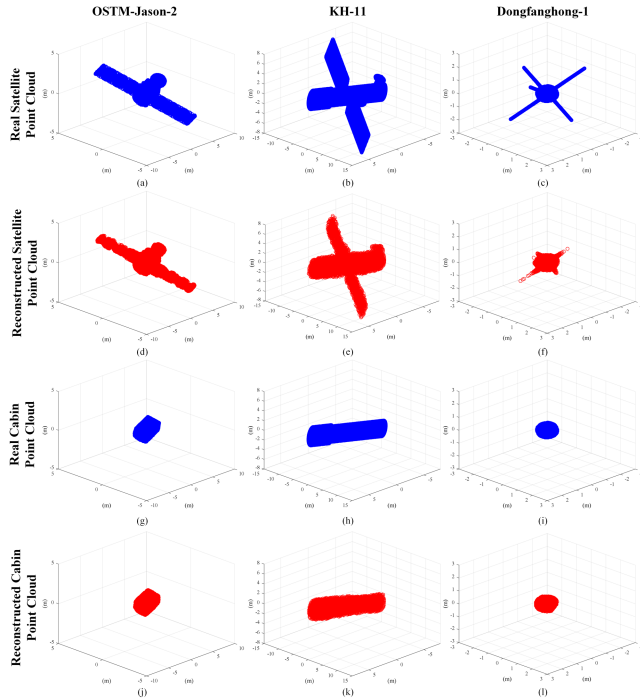


Fig. 15. Satellite reconstruction results and cabin reconstruction results. (a)–(c) Real satellite point clouds. (d)–(f) Reconstructed satellite point clouds. (g)–(i) Cabin point clouds manually segmented from real satellite point clouds. (j)–(l) Reconstructed cabin point clouds.

DPE, and the images in the second column are the ISAR images and target projection results with DPE. Our algorithm effectively corrects target deviations. To measure the accuracy of the DPE, an index dubbed the average key point distance d_{kp} is defined as follows.

Step 1: Extract the key points from the real point cloud. For OSTM-Jason-2 and KH-11, the vertices of their solar

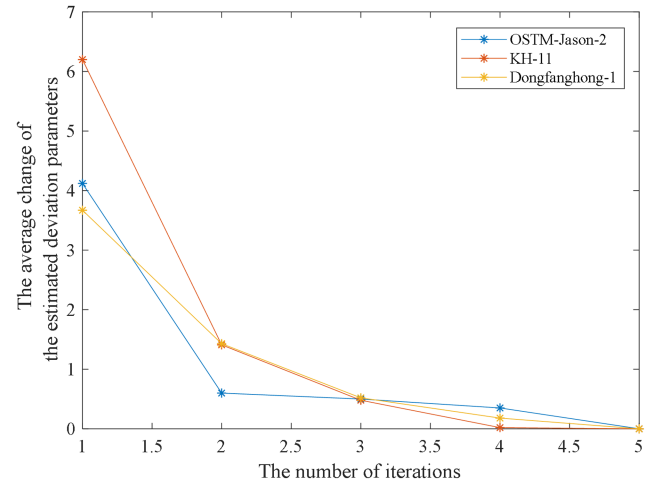


Fig. 16. Convergence of the satellite 3-D reconstruction with DPE.

planes are chosen as key points. For Dongfanghong-1, the centroid of its cabin is chosen as the key point.

Step 2: Calculate the projection $\vec{pk}_{i,n}$ of the i th key point on the n th image.

Step 3: Extract the key points $\vec{ek}_{i,n}$ from the n th image.

Step 4: Calculate the average key point distance as follows:

$$d_{kp} = \frac{1}{N} \frac{1}{\text{Inum}} \sum_{n=1}^N \sum_{i=1}^{\text{Inum}} \left\| \vec{pk}_{i,n} - \vec{ek}_{i,n} \right\|_2 \quad (19)$$

where the number of key points Inum is 1 or 4.

The variation curves of the average key point distance with respect to the number of iterations are given in Fig. 18. In Figs. 17 and 18, it is evident that, before registration, the targets in the ISAR images deviate from their theoretical projections, with an average deviation of approximately 0.2–0.4 m. After registration, the targets in the ISAR images

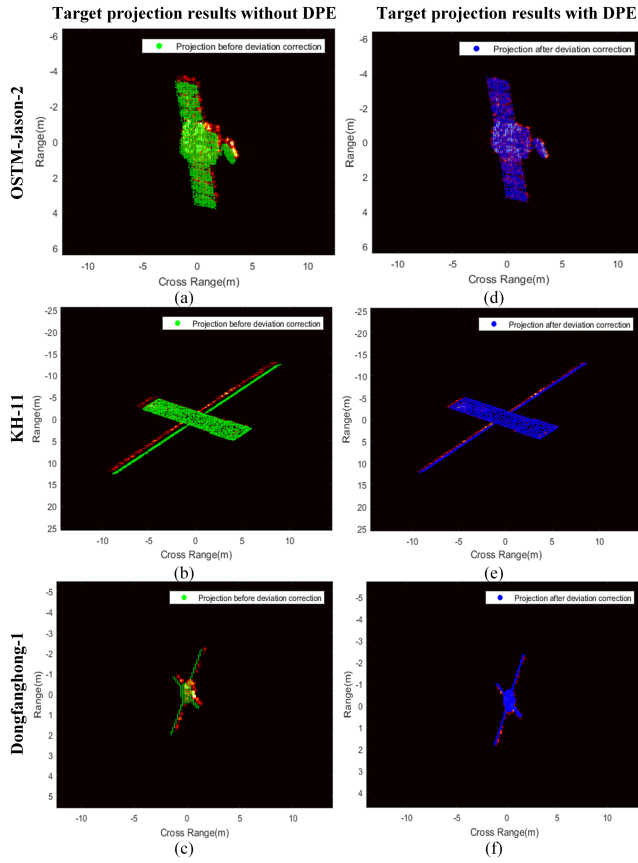


Fig. 17. Some deviation registration results. (a)–(c) Target projection results without DPE. (d)–(f) Target projection results with DPE.

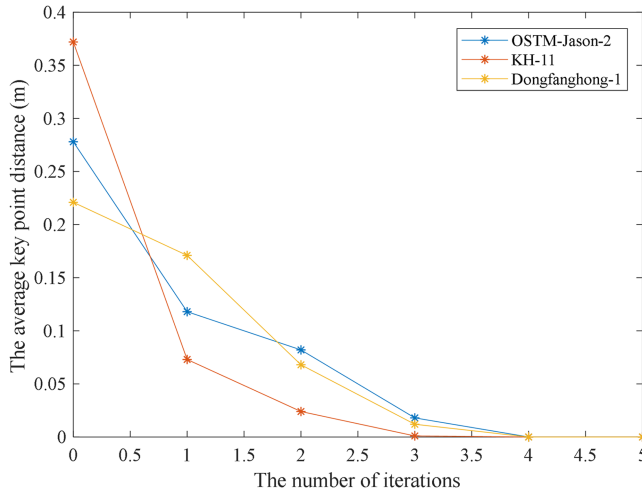


Fig. 18. Iteration curves of the average key point distance.

coincide with their theoretical projection locations. These results validate the necessity of DPE and confirm the accuracy of our DPE.

3) *Reliability of the CC*: Some CC results are shown in Fig. 19. Our CC module accurately completes the occluded cabins. Notably, the cabin of Dongfanghong-1 need not be completed as it is not occluded by other components.

Then, an index named occlusion-completion intersection over union OCIOU_n is defined by (20) to measure the

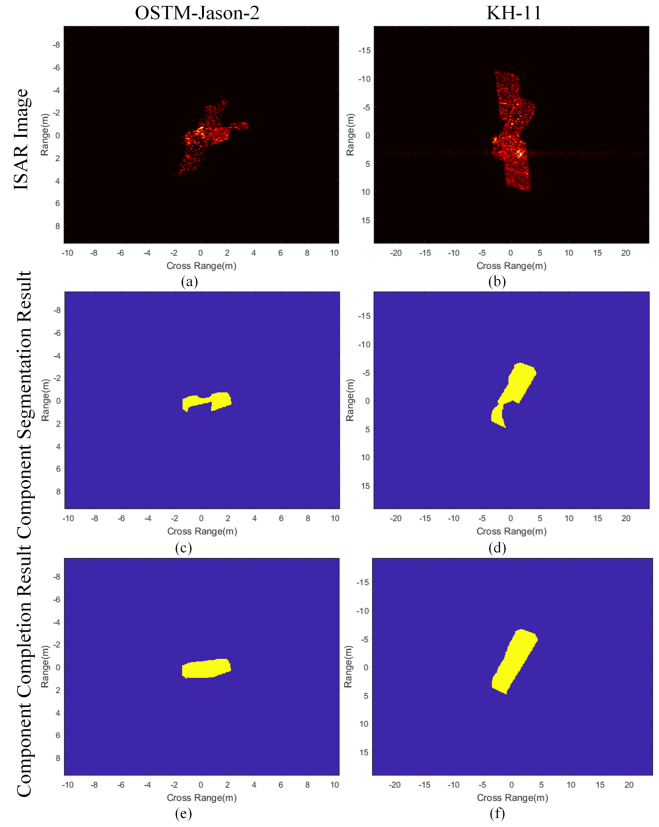


Fig. 19. Some CC results. (a) and (b) ISAR images. (c) and (d) Component segmentation results. (e)–(f) CC results.

TABLE IV
Mean OCIOUs for OSTM-Jason-2 and KH-11

Satellite	OSTM-Jason-2	KH-11
Mean OCIOU	0.65	0.88

reliability of our CC

$$\text{OCIOU}_n = \frac{\text{Area}_{r,n} \cap \text{Area}_{c,n}}{\text{Area}_{r,n} \cup \text{Area}_{c,n}} \quad (20)$$

where $\text{Area}_{r,n}$ is the real occluded part in the n th image, and $\text{Area}_{c,n}$ is the completed part. The mean OCIOU values for OSTM-Jason-2 and KH-11 are shown in Table IV. Their mean OCIOUs are higher than 0.65, which means that our CC module is reliable.

4) *Ablation Study*: In the ablation study, the RA is measured as follows:

$$\text{RA} = \frac{1}{3} \sum_{j=1}^3 \frac{\text{Bset}_{rc,j} \cap \text{Bset}_{CAD,j}}{\text{Bset}_{rc,j} \cup \text{Bset}_{CAD,j}} \quad (21)$$

where $\{\text{Bset}_{rc,j}\}$ are the three views of the reconstructed point cloud, and $\{\text{Bset}_{CAD,j}\}$ are the three views of the actual CAD model. The unions and intersections can be calculated by Margalit and Knott [43]. The RAs for satellite and cabin are provided in Table V. As bolded in Table V, the highest RA is obtained by the reconstruction with both DPE module and CC module. For all three satellites, the DPE module obviously improves the reconstruction performance, and the CC module also brings some improvement. The impact of

TABLE V
Ablation Study for the 3-D Geometry Reconstruction

Evaluation Index	CC Module	DPE Module	OSTM-Jason-2	KH-11	Dongfanghong-1
Satellite RA	N/A	×	0.8032	0.8068	0.4430
	N/A	√	0.8355	0.8531	0.5524
Cabin RA	×	×	0.8899	0.8660	0.8989
	×	√	0.9031	0.8943	0.9258
	√	×	0.8959	0.8744	0.8989
	√	√	0.9061	0.9007	0.9258

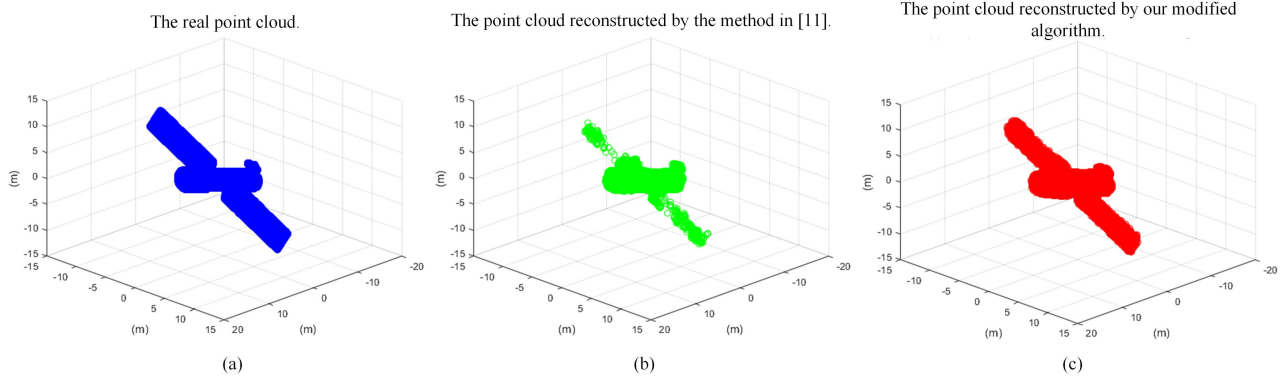


Fig. 20. Satellite reconstruction results when deviations exist in all the images. (a) Real point cloud. (b) Reconstruction results without DPE. (c) Reconstruction results with DPE.

TABLE VI
Computation Complexity and Time Consumption of the Reconstruction Process With and Without DPE

Reconstruction without DPE	Computation Complexity
	$11 \cdot N \cdot N_c$
	Mean Time Consumption
Reconstruction with DPE	12.462 s
	Computation Complexity
	$(11 + 2(2d_{r,\max} + 1) \cdot (2d_{a,\max} + 1) + K) \cdot N \cdot N_c$
	Mean Time Consumption
	232.658 s

the CC module is limited because there are few images where the cabin is almost complete.

It is worth noting that some small satellite components, such as antennas, cannot be effectively reconstructed due to image resolution and quality limitations, resulting in lower satellite RAs than cabin RAs.

Moreover, our reconstruction algorithm can still achieve satisfying reconstruction results even when deviations exist in all images. Fig. 20 shows the reconstruction results when deviations are added to all the images. As shown in Fig. 20(c), our DPE is still effective under these conditions.

5) *Computational Complexity Analysis*: Table VII presents the computational complexity of the reconstruction process with and without DPE. The mean time consumption of the three targets, which is obtained by running MATLAB codes on a personal computer with an Intel Core i7 3.40 GHz processor and 16 GB of memory, is also given in Table VI. where N is the number of images, N_c is the number of candidate points, $d_{r,\max}$ and $d_{a,\max}$ are the maximum deviation parameters, and T is the iteration number. In our

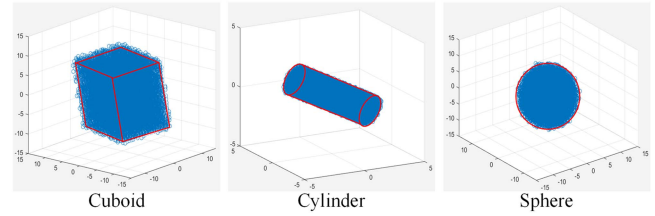


Fig. 21. Diagram of the generated point clouds.

experiments, $d_{r,\max} = 12$, $d_{a,\max} = 2$, and the mean value of T is 6. Although our computational complexity and time consumption are approximately 20 times those of the reconstruction without DPE, the mean satellite RA of all satellites is obviously improved. As shown in the first and second rows in Table V, their mean value improves from 0.6843 to 0.7470.

C. Performance of the Proposed 3-D Geometry Fitting Algorithm

In this section, the accuracy of our 3-D geometry fitting algorithm is proven via multiple Monte Carlo experiments on randomly generated point clouds. First, we randomly generated spherical, cylindrical, and cubic point clouds. For each shape, $\text{Num}_T = 100$ point clouds are generated, and their sizes and attitudes are different. To imitate the influence of image noise, we introduce slight random movements to the generated point clouds. Three of the generated point clouds are shown in Fig. 21. Then, our algorithm is employed to realize 3-D geometry fitting for these point clouds, and three evaluation indices are defined to measure the accuracy.

TABLE VII
Accuracy of the Proposed 3-D Geometry Fitting Algorithm

	Cuboid	Cylinder	Sphere
Shape Discrimination Accuracy	92%	99%	100%
Size Estimation Error	3.35%	2.95%	3.85%
Attitude Estimation Error	0.343°	0.375°	/

The shape discrimination accuracy is defined as follows:

$$R_{\text{shape}} = \frac{\text{Num}_C}{\text{Num}_T} \quad (22)$$

where Num_C is the number of correctly discriminated point clouds.

After excluding the points incorrectly distinguished, the size estimation accuracy is defined as follows:

$$E_{\text{size}} = \frac{1}{\text{Num}_C} \sum_{i=1}^{\text{Num}_C} \frac{\sum_{k=1}^K |l_{k,t,i} - l_{k,e,i}|}{\sum_{k=1}^K l_{k,t,i}}. \quad (23)$$

For cubic point clouds, $K = 3$ and $l_{k,t,i}$ are the actual lengths of the three orthogonal sides, and $l_{k,e,i}$ are the estimated lengths. For cylindrical point clouds, $K = 2$, $l_{k,t,i}$ and $l_{k,e,i}$ are the lengths of the main axis and bottom radius, respectively. For spherical point clouds, $K = 1$, $l_{k,t,i}$ and $l_{k,e,i}$ are the lengths of the radii.

The attitude estimation accuracy is defined as follows:

$$E_{\text{attitude}} = \frac{1}{\text{Num}_c} \sum_{i=1}^{\text{Num}_c} \theta_i. \quad (24)$$

For cubic point clouds, θ_i is the mean angle between the actual orientations and estimated orientations of the three orthogonal sides. For cylindrical point clouds, θ_i is the angle between the actual orientation and the estimated orientation of the main axis.

The experimental results are shown in Table VII. The shape discrimination operator we designed is robust, and our 3-D geometry fitting algorithm can estimate the size and attitude of a 3-D point cloud accurately. The three-view shape discrimination results for two ellipses and one rectangle are obtained in our experiment. In response to these discrimination errors, we rejudge the ellipse with a smaller operator value as a rectangle. By this strategy, some shape discrimination errors of cylindrical point clouds are rectified, resulting in an improvement in accuracy from 94% to 99%.

D. Performance of the Proposed Geometric Feature Extraction Framework

In this section, the existing method based on 2-D PCA [8] is taken as a comparison method to demonstrate the accuracy and wide applicability of our framework. Moreover, we analyzed the impact of our CC and DPE modules on geometric feature extraction via an ablation study. As shown in Table VIII, our methods with different modules are numbered from 1 to 4. For each method, a set of geometric parameters is estimated with all the simulated images. The

TABLE VIII
Numbering of Our Methods in the Ablation Study

	CC module	DPE module
Our method 1	×	×
Our method 2	√	×
Our method 3	×	√
Our method 4	√	√

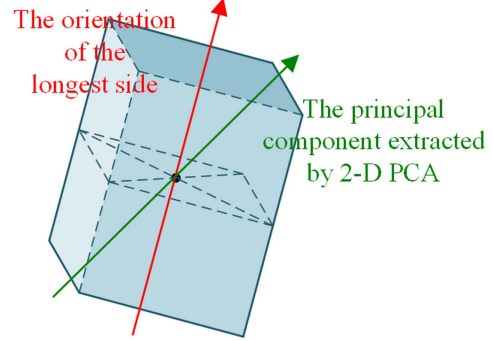


Fig. 22. Relationship between the longest side of the cuboid and the principal component of the projection.

best estimation results among these methods are bolded in following Tables.

1) *Results of OSTM-Jason-2*: The geometric feature extraction results of OSTM-Jason-2 are shown in Table IX. The 3-D size and absolute attitude of a cabin can be described by the lengths and orientations of three orthogonal sides. The existing method based on 2-D PCA can only estimate the length and orientation of the longest side of the cabin, while our method can discriminate the 3-D shape of the cabin automatically and estimate the lengths and orientations of three sides accurately. Moreover, our estimation accuracy is much better. The size and orientation estimation error of the existing method are more than twice ours. This is largely because 2-D PCA cannot accurately extract the projections of the longest cabin side in most images. As shown in Fig. 22, the projection of a cubic cabin is not axisymmetric, and the longest side is inconsistent with the extracted principal component; moreover, the reconstructed 3-D point cloud in our method is axisymmetric, and the orientation of the longest side is proven to be available by 3-D PCA.

In the ablation study, the DPE module improved the size estimation accuracy by almost 2% and the attitude estimation accuracy by almost 2°. The CC module can also improve the estimation accuracy slightly in most cases. However, when the signal sidelobes in images are mistaken for the target pixels, the length estimation result will be larger than the true value, and the CC module may further increase the error. In Table IX(c), for length estimation for side 2, the method with only the DPE module achieved the highest accuracy. Moreover, even without these two modules, our method achieves higher accuracy than the existing method does, indicating that our geometric fitting algorithm is robust to RA.

TABLE IX
Geometric Feature Extraction Result of OSTM-Jason-2

(a) Shape Discrimination Result

True value	Cuboid
2-D PCA method	Fail
Our methods	Cuboid

(b) Size and Orientation Estimation Result of Side 1

	Length		Orientation	
	Result	Error	Result	Error
True value	3.17 m	N/A	(0.940,0.000,0.342)	N/A
2-D PCA method	3.41 m	0.24 m	(0.975,0.082,0.208)	9.19°
Our method 1	3.29 m	0.12 m	(0.943,-0.025,0.331)	1.59°
Our method 2	3.29 m	0.12 m	(0.943,-0.024,0.333)	1.45°
Our method 3	3.25 m	0.08 m	(0.941,-0.009,0.338)	0.56°
Our method 4	3.25 m	0.08 m	(0.941,-0.003,0.338)	0.30°

(c) Size and Orientation Estimation Result of Side 2

	Length		Orientation	
	Result	Error	Result	Error
True value	1.37 m	N/A	(-0.089,0.966,0.243)	N/A
2-D PCA method	Fail			
Our method 1	1.50 m	0.13 m	(-0.085,0.946,0.313)	4.16°
Our method 2	1.57 m	0.20 m	(-0.058,0.980,0.189)	3.67°
Our method 3	1.42 m	0.05 m	(-0.076,0.954,0.289)	2.83°
Our method 4	1.44 m	0.07 m	(-0.065,0.966,0.252)	1.44°

(d) Size and Orientation Estimation Result of Side 3

	Length		Orientation	
	Result	Error	Result	Error
True value	1.37 m	N/A	(-0.330,-0.259,0.908)	N/A
2-D PCA method	Fail			
Our method 1	1.35 m	0.02 m	(-0.321,-0.323,0.890)	3.86°
Our method 2	1.37 m	0.00 m	(-0.333,-0.197,0.922)	3.63°
Our method 3	1.37 m	0.00 m	(-0.323,-0.298,0.898)	2.37°
Our method 4	1.37 m	0.00 m	(-0.328,-0.259,0.909)	0.48°

TABLE X
Geometric Feature Extraction Result of KH-11

	Shape	Main axis length		Main axis orientation		Bottom radius length	
		Result	Error	Result	Error	Result	Error
True value	Cylinder	12.00 m	N/A	(0.955,-0.211,0.210)	N/A	1.77 m	N/A
2-D PCA method	Fail	12.24 m	0.24 m	(0.942,-0.241,0.233)	2.31°	Fail	
Our method 1	Cylinder	11.63 m	0.37 m	(0.943,-0.236,0.236)	2.16°	1.82 m	0.05 m
Our method 2		11.68 m	0.32 m	(0.943,-0.236,0.236)	1.67°	1.84 m	0.07 m
Our method 3		11.78 m	0.22 m	(0.943,-0.236,0.236)	1.67°	1.82 m	0.05 m
Our method 4		11.83 m	0.17 m	(0.953,-0.197,0.230)	1.43°	1.82 m	0.05 m

2) *Results of KH-11*: The geometric feature extraction results for KH-11 are shown in Table X. The 3-D size and absolute attitude of its cabin can be described by the length and orientation of the main axis together with the bottom radius. The existing method based on 2-D PCA [8] can only estimate the length and orientation of the main axis, while our method can additionally discriminate the cabin shape and estimate the bottom radius. Moreover, compared with the existing method based on 2-D PCA [8], our attitude and size estimation errors are slightly lower. In the ablation study, the size and attitude estimations with both the CC module and DPE module are the most accurate.

3) *Results of Dongfanghong-1*: The geometric feature extraction results for Dongfanghong-1 are shown in Table XI. For a spherical satellite, the key aspect of its

TABLE XI
Geometric Feature Extraction Result of Dongfanghong-1

	Shape	Radius length	
		Result	Error
True value	Sphere	0.59 m	N/A
2-D PCA method	Fail		
Our method 1	Sphere	0.58 m	0.01 m
Our method 2		0.58 m	0.01 m
Our method 3		0.59 m	0.00 m
Our method 4		0.59 m	0.00 m

geometric feature extraction is shape discrimination. Once the shape is determined, the radius can be easily estimated. The existing method based on 2-D PCA has no shape discrimination ability. As shown in Table XI, our method can discriminate shape and estimate radius accurately.

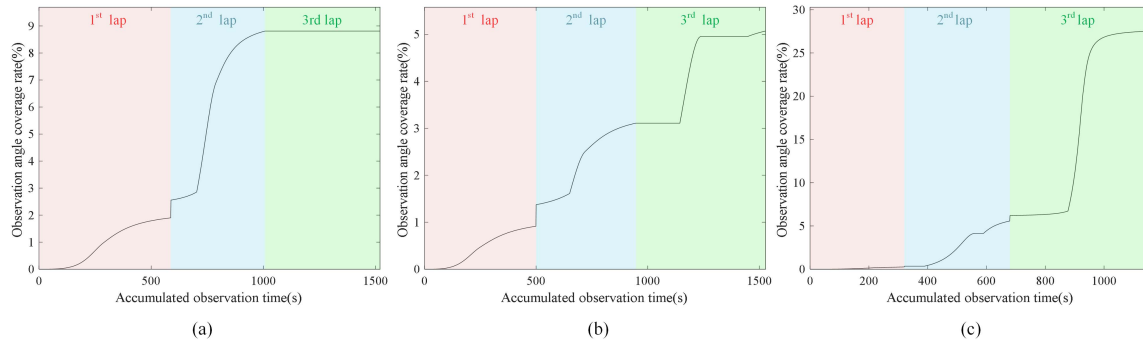


Fig. 23. Variation curves of the observation angle coverage rates. (a) Observation angle coverage rate of OSTM-Jason-2. (b) Observation angle coverage rate of KH-11. (c) Observation angle coverage rate of Dongfanghong-1.

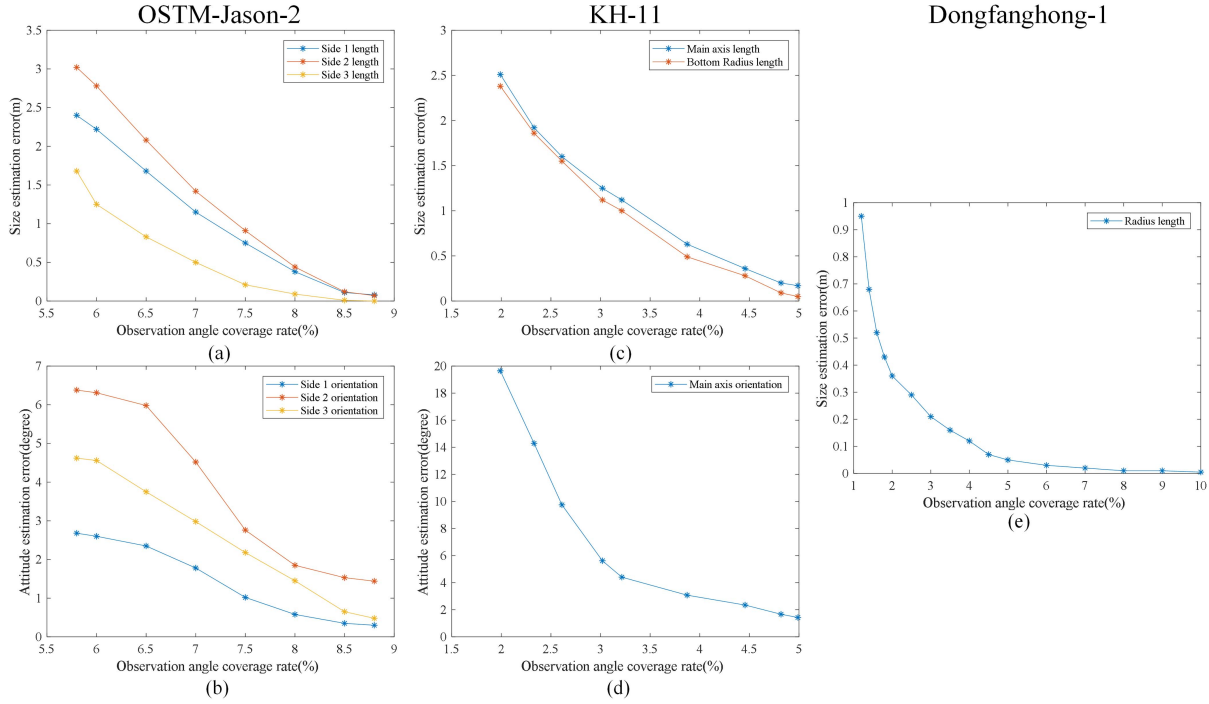


Fig. 24. Variation curves of the attitude and size estimation errors with increasing observation angle. (a) Attitude estimation error of OSTM-Jason-2. (b) Size estimation error of OSTM-Jason-2. (c) Attitude estimation error of KH-11. (d) Size estimation error of KH-11. (e) Size estimation error of Dongfanghong-1.

4) *Effect of Observation Angle Richness on Estimation Accuracy:* The observation angle richness has a significant impact on the estimation accuracy. Here, an index named the observation angle coverage rate is defined by (25) to measure the observation angle richness

$$\text{rate}_{p-c} = \frac{\theta_{dAz}}{360^\circ} \cdot \frac{\theta_{dEl}}{180^\circ} \quad (25)$$

where θ_{dAz} is the maximum difference between the azimuth angles, and θ_{dEl} is the maximum difference between the elevation angles. The variation curves of the observation angle coverage rates are shown in Fig. 23.

Variation curves of our estimation errors with respect to the observation angle richness are shown in Fig. 24.

In Fig. 24(a) and (b), the accurate shape discrimination for OSTM-Jason-2 is achieved when rate_{p-c} exceeds 5.8%. Additionally, the attitude and size estimation errors for the three sides of OSTM-Jason-2 remain below 0.2 m (with a 10% error) and 5° when rate_{p-c} exceeds 8.5%. In Fig. 24(c) and (d), accurate shape discrimination for KH-11 is achieved when rate_{p-c} exceeds 2%. In addition, the attitude and size estimation errors for the main axis of KH-11 remain below 1.2 m and 5° when rate_{p-c} exceeds 3%. Additionally, the bottom-size estimation error remains below 0.18 m when rate_{p-c} exceeds 4.8%. Based on the above analysis, more abundant observation angles are required for the accurate estimation of the 3-D size and attitude than for the estimation of the main axis. In Fig. 24(e), the accurate shape discrimination of Dongfanghong-1 is achieved when rate_{p-c}

TABLE XII
Results of Geometric Feature Extraction From the Measured Data

	Shape	Main axis length		Main axis orientation		Bottom radius length	
		Result	Error	Result	Error	Result	Error
True value	Cylinder	10.40 m	N/A	(0.984,0.169,0.047)	N/A	1.90 m	N/A
2-D PCA method	Fail	9.98 m	0.42 m	(0.992,0.129,-0.018)	4.76°	Fail	
Our method 1	Cylinder	9.97 m	0.43 m	(0.986,0.161,-0.026)	4.57°	2.22 m	0.32 m
Our method 2		10.05 m	0.35 m	(0.987,0.161,-0.022)	4.36°	2.22 m	0.32 m
Our method 3		10.08 m	0.32 m	(0.986,0.165,-0.011)	3.73°	2.18 m	0.28 m
Our method 4		10.12 m	0.28 m	(0.986,0.168,-0.011)	3.53°	2.18 m	0.28 m

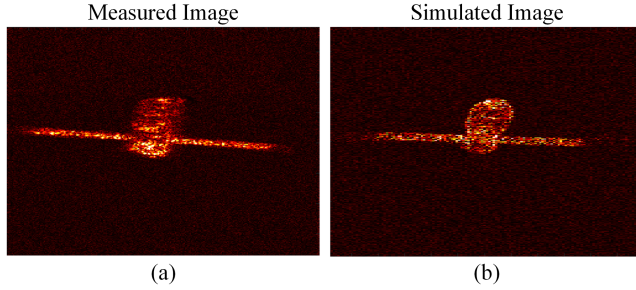


Fig. 25. Measured and simulated images of Tiangong-1. (a) Measured image. (b) Simulated image.

exceeds 1.25%, and its size estimation error remains below 0.06 m when rate_{p-c} exceeds 5%.

5) *Results on Measured Data:* We carry out experiments on the measured ISAR images of Tiangong-1. The imaging parameters are consistent with those in Table II, and the observation angle richness is 9.2%. As shown in Fig. 25, the measured and simulated images are visually similar, which indicates that our methods can be applied in reality. The experimental results are given in Table XII. The bottom size cannot be estimated by the existing method, and the estimation error obtained by our method is 0.28 m. For the main axis, the estimation accuracies of our framework are 0.14 m and 1.23° better than those of the existing method. Our method outperforms the existing methods in realistic scenarios.

To sum up, in simulation experiments on three different satellites, our framework extracts more diverse geometric features than does the existing method. For a cuboid cabin, we can estimate the size and direction of three orthogonal sides, while the existing method can only estimate the longest side. For a cylindrical cabin, we can estimate the size and direction of its main axis together with its bottom size, while the existing method can only estimate its main axis. For a spherical cabin, we can estimate its radius; however, the existing methods are not applicable. Moreover, in realistic scenarios, our framework consistently outperforms the existing method.

E. Some Discussions

The experimental results above illustrate the reliability and superiority of our methods. However, there are still some certain limitations that may affect the practical applications of our framework.

First, the accurate reconstruction requires massive amounts of observation data from different views. Our experiments utilize three-lap observation data from one single sensor to ensure diverse views, but this approach is costly. In practice, the multisensor fusion strategy can be adopted to reduce the observation time. For example, optical-and-radar image fusion for 3-D reconstruction can be realized based on the physical model introduced in [15].

In addition, geometric feature extraction is based on image interpretation, which presents challenges in satellite component segmentation due to the complexity and diversity of satellite structures. Most existing intelligent segmentation methods rely on extensive training data and time-consuming human labeling. These methods can hardly perform well on new targets with significant differences in structure. The generalization capability of component segmentation methods limits the application of geometric feature extraction methods.

Moreover, our CC module and 3-D geometric fitting algorithm assume that the satellite cabin is either a sphere, a quasi-cylinder, or a cuboid. As shown in Table I, many cabin satellites can be regarded as having these shapes. In addition, some satellites have additional loads attached to their cabins. Large loads are removed during the component segmentation process, and small loads do not significantly alter the cabin shapes in images due to limitations in image resolution. Additionally, some other satellites, such as octagonal prisms, have relatively complex cabins. However, due to the limited image resolution, projections of these cabins in images closely resemble those of cylinders, and they can also be approximated as quasi-cylinders. Therefore, the majority of satellite cabin shapes can be categorized as spheres, cylinders, or cuboids, suggesting the applicability of our method to most satellites. However, there are still a few satellites with unique cabins for which a more adaptable 3-D geometric fitting algorithm is needed.

V. CONCLUSION

In this article, a geometric feature extraction framework based on 3-D geometry reconstruction is proposed to estimate the 3-D shape, 3-D size, and absolute attitude of a satellite cabin. With a CC module to address the CC and a DPE module to eliminate the target deviations, we rebuild the 3-D cabin structure accurately. In view of diverse and unknown cabin shapes, we propose a 3-D geometry fitting algorithm applicable to different shapes to extract features,

in which the fitting objective functions are designed based on automatic 3-D shape discrimination. The experimental results validate the effectiveness of our methods. In comparison with the existing methods, our framework can extract more diverse geometric features and is applicable to satellites with different shapes. However, considering the real-time space situational awareness requirements, the application of these methods with limited observation data needs further investigation.

APPENDIX I MATHEMATICAL DERIVATION FOR THE 3-D PCA ON A REFLECTION SYMMETRICAL CABIN

Points of a reflection symmetrical cabin are distributed symmetrically around a plane. This plane is called the reflection symmetrical plane. In a coordinate system with its z -axis aligned to the normal vector of the reflection symmetrical plane, the x coordinates and y coordinates of points above the plane are consistent with those of points below the plane, and their z coordinates are opposite.

In any coordinate system, a 3-D reflective symmetric point cloud, which is uniformly distributed, can be represented as follows:

$$P = \left\{ \mathbf{R} \begin{bmatrix} x_i \\ y_i \\ z_i \end{bmatrix} \middle| \begin{bmatrix} x_j \\ y_j \\ z_j \end{bmatrix} = \begin{bmatrix} x_{j+K} \\ y_{j+K} \\ -z_{j+K} \end{bmatrix} \right\} \quad (26)$$

where $\mathbf{R}$ is a rotation matrix, $i = 1, 2, \dots, 2K$, and $j = 1, 2, \dots, K$. The normal vector of its reflective symmetry plane is $\mathbf{R}[0, 0, 1]^T$.

The components extracted by the PCA are eigenvectors of the covariance matrix. The covariance matrix can be calculated as follows:

$$\mathbf{C} = \frac{1}{2K} \bar{\mathbf{W}} \bar{\mathbf{W}}^T \quad (27)$$

where $\bar{\mathbf{W}}$ is the centralized location matrix. Considering the centroid of this point cloud is $[0, 0, 0]^T$, $\bar{\mathbf{W}}$ can be calculated as follows:

$$\bar{\mathbf{W}} = \mathbf{R} \begin{bmatrix} x_1 & \dots & x_i & \dots & x_{2K} \\ y_1 & \dots & y_i & \dots & y_{2K} \\ z_1 & \dots & z_i & \dots & z_{2K} \end{bmatrix}. \quad (28)$$

Then, its covariance matrix can be represented as follows:

$$\mathbf{C} = \frac{1}{2K} \mathbf{R} \mathbf{D}_c \mathbf{R}^T \quad (29)$$

$$\mathbf{D}_c = \begin{bmatrix} d_{11} & d_{12} & d_{13} \\ d_{21} & d_{22} & d_{23} \\ d_{31} & d_{32} & d_{33} \end{bmatrix} \quad (30)$$

where $d_{11} = 2 \sum_{j=1}^K x_j^2$, $d_{22} = 2 \sum_{j=1}^K y_j^2$, $d_{12} = d_{21} = 2 \sum_{j=1}^K x_j y_j$, $d_{13} = d_{31} = \sum_{j=1}^K x_j (z_j - z_j) = 0$, $d_{23} = d_{32} = \sum_{j=1}^K y_j (z_j - z_j) = 0$, and $d_{33} = 2 \sum_{j=1}^K z_j^2$.

The rotation matrix $\mathbf{R}$ is an orthogonal matrix, i.e., $\mathbf{R} \mathbf{R}^T = \mathbf{I}$. Therefore, we can obtain the following equation:

$$\begin{aligned} \mathbf{C} \begin{pmatrix} \mathbf{R} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \end{pmatrix} &= \frac{1}{2K} \mathbf{R} \mathbf{D}_c \mathbf{R}^T \begin{pmatrix} \mathbf{R} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \end{pmatrix} \\ &= \frac{1}{2K} \mathbf{R} \mathbf{D}_c \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \\ &= \frac{\sum_{i=1}^K z_{c,i}^2}{K} \mathbf{R} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. \end{aligned} \quad (31)$$

According to (31), the normal vector $\mathbf{R}[0, 0, 1]^T$ is one of the eigenvectors of the covariance matrix. This means that the normal vector of the reflective symmetry plane can be available by the 3-D PCA.

APPENDIX II MATHEMATICAL DERIVATION FOR THE MINIMUM DISTANCE BETWEEN A POINT AND AN ELLIPSE

The minimum distance between a point (x_p, y_p) and an ellipse can be obtained by solving the following optimization:

$$\begin{aligned} d_{\min} &= \min_{x,y} \sqrt{(x - x_n)^2 + (y - y_n)^2} \\ \text{s.t. } &\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \end{aligned} \quad (32)$$

where $\begin{bmatrix} x_n \\ y_n \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \left(\begin{bmatrix} x_p \\ y_p \end{bmatrix} - \begin{bmatrix} x_c \\ y_c \end{bmatrix} \right)$, $2a$ is the length of the long axis, and $2b$ is the length of the short axis, and θ is the inclination angle of the ellipse, $[x_c, y_c]^T$ is the location of the ellipse centroid, and $[x_p, y_p]^T$ is the location of the point.

The optimization can be solved by the Lagrange multiplier algorithm. The optimization can be converted into the following optimization:

$$F(x, y, \lambda) = (x - x_n)^2 + (y - y_n)^2 + \lambda \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 \right) \quad (33)$$

where λ is the multiplier. Then, the best $[x, y]^T$ can be obtained by solving the following equations:

$$\begin{cases} \frac{\partial F}{\partial x} = 2(x - x_n) + 2\lambda \frac{x}{a^2} = 0 \\ \frac{\partial F}{\partial y} = 2(y - y_n) + 2\lambda \frac{y}{b^2} = 0 \\ \frac{\partial F}{\partial \lambda} = \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 = 0. \end{cases} \quad (34)$$

The best x can be calculated by $x = x_n a^2 / (a^2 + \lambda_o)$, the best y can be calculated by $y = y_n b^2 / (b^2 + \lambda_o)$, and the minimum distance can be calculated as follows:

$$d_{\min} = \sqrt{\left(\frac{\lambda_o}{a^2 + \lambda_o} x_n \right)^2 + \left(\frac{\lambda_o}{b^2 + \lambda_o} y_n \right)^2} \quad (35)$$

where λ_o is the root of the quaternion (34), which can be solved by Ferrari's formula

$$\lambda^4 + A\lambda^3 + B\lambda^2 + C\lambda + D = 0$$

$$\begin{cases} A = 2(a^2 + b^2) \\ B = (a^4 + b^4 + 4a^2b^2 - x_n^2a^2 - y_n^2b^2) \\ C = 2(a^2b^4 + a^4b^2 - a^2b^2(x_n^2 + y_n^2)) \\ D = a^4b^4 - x_n^2a^2b^4 - y_n^2a^4b^2. \end{cases} \quad (36)$$

Considering that the quaternion (35) may have more than one root, the root corresponding to the minimum distance is chosen as the best $[x, y]^T$.

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